Characteristic-order pairings: actual domains and failed strengthenings

Result. For every prime p, characteristic-p field and natural k, the only p^k-th unit root is 1. Over every finite such field, the unit-power map is surjective and the actual quotient by its image has cardinality one. A left-bilinear map from a p^k-killed additive group into any characteristic-p field's units is identically one. A nonzero left element precludes nondegeneracy. [pcc:unverified]

Any char-p field	p^k-th unit roots: identity only Finite field: units / p^k-th powers has cardinality 1
Actual curve domains	Finite E/nE and E[n] have equal cardinality Ordinary char2: 2 / 2 / 1; paired ordinary char3: 3 / 3 / 1
Scope controls	Prime nonidentity root implies coprimality Imperfect F2(X): X gives a nonidentity square class

Actual domains. Lean proves for every finite additive commutative group that the quotient by n-multiples has the same cardinality as the actual n-kernel, then specializes to Mathlib curve points. Left domain E[n], right domain E/nE and target K*/(K*)^n are actual kernels/ranges/quotients. Frobenius injectivity, finite surjectivity and the subgroup-index theorem supply the proofs, with no candidate pairing or nondegeneracy premise. [pcc:unverified]

Unbounded curve families. Every finite characteristic-two field and nonsingular general model has left/right2-kernel/cokernel cardinality1 if a1=0 and2 otherwise; target cardinality1. Every ordinary model excludes unit-target candidates over any characteristic-two field and candidates on the proper Tate domains. The actual $y^2+xy=x^3+1$ model has point(0,1) of exact order2 over every such field. In finite characteristic3, a constructed nonzero3-point on one ordinary cubic or its nonsquare partner gives actual domain sizes3/3/1. The credited F3 partner has P=(1,1) of exact order3. Prior actual models, geometric torsion, twist construction and group proofs are credited. [pcc:unverified]

Source guard. Galbraith author chapter26, Definition26.2.2/Theorem26.2.3 and Exercise26.3.1/Theorem26.3.3, printed574–577/PDF2–5 read and visually inspected: coprimality and Tate root/domain hypotheses retained. Lean proves a prime-order nonidentity root already forces coprimality with field cardinality. Direct universal negations concern the unit claim after deleting coprimality and the separate unqualified Tate claim omitting both coprimality and root hypotheses. They do not refute the retained-root-qualified source or perfect group-scheme pairings. [pcc:unverified]

Imperfect-field negative control. Over F2(X), the actual X-unit has a nonidentity class modulo squares; target cardinality is not1 and the square map is not surjective. The prior integer-degree nonsquare proof is credited. Unit square-roots of1 remain trivial. Lean directly refutes deleting finiteness from the universal trivial-target claim. [pcc:unverified]

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